# Data Mining COSC3126
# Assignment 1-1 Practical Data Mining

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| Assessment Type | This is an individual assignment, meaning that you must complete this assignment by yourself. You must submit your solution online via Canvas.<br>Clarifications or updates may be made via announcements or relevant discussion forums. |
| Due Date | 28 March 2026, 17:59 PM |
| Marks | 20 |

## 1 Overview

In this assignment you will work on typical tasks in practical data mining. You will apply a number of methods to given data sets and write a report on your findings. Your assignment will be assessed on demonstrated understanding of concepts, algorithms, methodology, analysis of results and conclusions. Marks are awarded for meeting requirements as closely as possible (see rubrics). Please make sure your answers are labelled correctly with the corresponding part and sub-question numbers, to make it easier for the marker to follow. Please stick to the required page limits (penalty may apply).

## 2 Learning Outcomes

This assessment relates to the following learning outcomes of the course.

- CLO 1: Demonstrate advanced knowledge of data mining concepts and techniques.
- CLO 2: Apply the techniques of clustering, classification, association finding, feature selection and visualization on real world data.
- CLO 4: Apply data mining software and toolkits in a range of applications.
- CLO 5: Set up a data mining process for an application, including data preparation, modelling and evaluation.

## 3 Assignment Details

You will perform the following tasks using Weka software on the data sets given with this assignment specification (unless otherwise stated). You may need to prepare the data file in **arff** format for using in Weka.

Reports must be in page size A4, font Times New Roman with size 12, standard margin.

## Task 1: Classification (12 marks)

This part of the assignment is concerned with the data file **chronic-kidney-disease.arff**. A description of the data is provided in the file *chronic-kidney-disease.info.txt*. The main goal here is to achieve the highest classification accuracy with the lowest amount of overfitting.

1. Run the following classifiers, with the default parameters, on the data: **ZeroR, OneR, J48, IBK** and construct a table of the training and cross-validation errors (see an example below). You can get the training error by selecting "Use training set" as the test option. What can you conclude from these results? Provide your explanation.

| Run No | Classifier | Parameters | Training Error | Cross-validation Error | Overfitting |
|---|---|---|---|---|---|
| 1 | ZeroR | None | 30.0% | 28.0% | None |
| . | . | . | . | . | |

2. Using the **J48** classifier, can you find a combination of the C and M parameter values that minimizes the amount of overfitting? Include the results of your best five runs, including the parameter values, in your table of results. What is your conclusion? Provide your explanation.
3. Reset **J48** parameters to their default values. What is the effect of lowering the number of examples in the training set? Provide your explanation. Include your runs in your table of results.
4. Using the **IBk** classifier, can you find the value of k that minimizes the amount of overfitting? Provide your explanation. Include your runs in your table of results.
5. Try two other classifiers. Aside from ZeroR, which classifiers are best and worst in terms of predictive accuracy? Include 5 runs in your table of results. Provide your analysis on these results.
6. Compare the accuracy of ZeroR, OneR and J48. What do you conclude? Give your explanation on these results.
7. How does model performance change when tested on different subgroups (e.g., age groups, gender, severity of disease)?
8. Use an attribute selection algorithm to get a reduced attribute set. How does the accuracy on the reduced set compare with the accuracy on the full set? Provide your explanation.

**Report Length**: Up to two pages, not including the table(s) of results.

## Task 2: Numeric Prediction (8 marks)

This task is concerned with prediction of the quality attribute in the **wine_quality** data set.

1. Run the following models, with default parameters, on this data: **ZeroR, M5P, IBk** and construct a table of training and cross-validation errors. You may want to turn on "Output Predictions" to get a better sense of the magnitude of the error on each example. What do you conclude from these results? Give your explanation.

2. Explore different parameter settings for M5P and IBk. Which values give the best performance in terms of predictive accuracy and overfitting? Include the results of the best five runs in your table of results. Provide your explanation on these results.

3. Investigate two other classifiers for numeric prediction and their associated parameters. Include your best five runs in your table of results. Which classifier gives the best performance in terms of predictive accuracy and overfitting? Provide your explanation.
   Let's consider the best classifier for this dataset. How does model performance change when tested on different types of wine (e.g., red vs. white wine)?

4. Use an attribute selection algorithm to get a reduced attribute set. How does feature selection impact the models' performance? Explain the feature engineering techniques that could improve numeric prediction accuracy for this task.

**Report Length**: Up to two pages, not including the table(s) of results.

## 4 Submission Instructions

You need to submit the following file(s) via Canvas:

- **one PDF file** for the report covering solutions for Task 1 and Task 2.

**Late submission penalty**

After the due date, you will have 5 business days to submit your assignment as a late submission. Late submissions will incur a penalty of 10% per day. After these five days, Canvas will be closed, and you will lose ALL the assignment marks.

**Assessment declaration**:
When you submit work electronically, you agree to the assessment declaration:
https://www.rmit.edu.au/ students/my-course/assessment-results/academic-integrity

## 5 Academic integrity and plagiarism

Academic integrity is about honest presentation of your academic work. It means acknowledging the work of others while developing your own insights, knowledge and ideas. You should take extreme care that you have:

- Acknowledged words, data, diagrams, models, frameworks and/or ideas of others you have quoted (i.e. directly copied), summarised, paraphrased, discussed or mentioned in your assessment through the appropriate referencing methods.
- Provided a reference list of the publication details so your reader can locate the source if necessary. This includes material taken from Internet sites, **AI tools**. If you do not acknowledge the sources of your material, you may be accused of plagiarism because you have passed off the work and ideas of another person without appropriate referencing, as if they were your own.

RMIT University treats plagiarism as a very serious offence constituting misconduct. Plagiarism covers a variety of inappropriate behaviours, including:

- Failure to properly document a source.
- Copyright material from the Internet, databases or AI tools.
- Collusion between students.

For further information on our policies and procedures, please refer to the following: https://www.rmit.edu.au/students/my-course/assessment-results/academic-integrity.

## 6 Marking guidelines

Factors contributing to the final mark will include the number of tasks attempted, the amount of exploration and demonstrated understanding of the algorithms, methodology, logical analysis, presentation of results and conclusions (see the marking rubrics in Canvas).